B01076645 || Florence-Sarah Egwu

Problem 2

Q1 Retrieve Names & GPA of all students in "Computer Science' major with GPA > 3.5

Ans Π Name, GPA (σ Major = 'Computer Science' $\wedge$ GPA > 3.5 (STUDENT))

Reason Π is Projection, it returns only values for the selected attributes i.e Name and GPA.
σ is Selection, looks into the STUDENT relation and returns all row that pass the criteria given. Since both condition must be true, $\wedge$ (intersect) is used to join the conditions

Q2 Return Name and Salary of Professors in 'Mathematics' department with the rank of 'Associate Professor' or higher

Ans Π Name, Salary (σ Department = 'Mathematics' $\wedge$ (Rank = 'Associate Professor' $\vee$ Rank = 'Full Professor') (PROFESSOR))

Reason Rank of Associate Professor or higher means we should return rows with a rank of Associate Professor OR Full Professor (this is higher), the $\vee$ (union) operator ensures this
σ gives all the professors in Maths department and ($\wedge$) whose rank meets our conditions.

Q3 List Course Title and Credit count of all courses at Level CSC14000 and above that carry > 3 credits

Ans Π Title, Credits (σ Level >= 'CSC14000 $\wedge$ Credits > 3) (COURSE))

Reason σ returns only tuples whose Level is at least CSC14000

and ($\wedge$) whose Credits is strictly greater than 3.
Π returns just the Title and Credits

Q4 Find Students who are either under 22 years old or have a GPA above 3.8. Return Name and Age

Ans Π Name, Age (σ Age < 22 $\vee$ GPA > 3.8 (STUDENT))

Reason σ returns Students who satisfy either one of the conditions, age < 22 OR GPA > 3.8, the $\vee$ (OR) is used to enforce that.
Π returns only Name and Age.